

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_STAR CO_SC (MODEL-A & APEX) **Date: 04-01-2025**

Time: 3 Hrs **GTM-11(THEORY BASED-1)** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A sphere cannot roll on

   A) a smooth horizontal surface B) a smooth inclined surface

   C) a rough horizontal surface D) a rough inclined surface

2. Given below are two statements

   Statement-I: The average momentum of a molecule in a sample of an ideal gas depends on temperature

   Statement-II: The rms speed of oxygen molecules in a gas is V. If the temperature is doubled and oxygen molecules dissociate into oxygen atoms, the rms speed will become 2V.

   In the light of the above statements, choose the correct option from the options given below.

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A) Both statements I and II are true.

B) Both statements I and II are false.

C) Statement-I is false but Statement-II is true.

D) Statement-I is true but Statement-II is false.

3. Two balls are thrown simultaneously in air. The acceleration of the centre of mass of the two balls while in air.

A) depends on the direction of the motion of the balls

B) is equal to g.

C) depends on the masses of the two balls

D) depends on the speeds of the two balls

4. Match List I with List II

	List I		List II
(a)	Magnetic Induction	(i)	$ML^2T^{-2}A^{-1}$
(b)	Magnetic Flux	(ii)	$M^0L^{-1}A$
(c)	Magnetic permeability	(iii)	$MT^{-2}A^{-1}$
(d)	Magnetization	(iv)	$MLT^{-2}A^{-2}$

Choose the most appropriate answer from the options given below:

A) a-ii, b-iv, c-i, d-iii B) a-ii, b-i, c-iv, d-iii

C) a-iii, b-ii, c-iv, d-i D) a-iii, b-i, c-iv, d-ii

5. Given below are two statements one is labeled as Statement-I: and the other is labeled as Statement-II:

Statement-I: The number of significant figures depends on the least count of measuring instrument.

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Statement-II: Significant figures depends on precision of measuring instrument.

In the light of the above statements, choose the most appropriate answer from the options given below:

A) Both statements are true.

B) Statement-I is true and Statement-II is false.

C) Statement-I is false and Statement-II is true.

D) Both statements are false.

6. Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R

Assertion (A): If dQ and dW represent the heat supplied to the system and the work done on the system respectively. Then according to the first law of thermodynamics $dQ = dU - dW$

Reason (R): First law of thermodynamics is based on law of conservation of energy.

In the light of the above statements, choose the correct answer from the option given below:

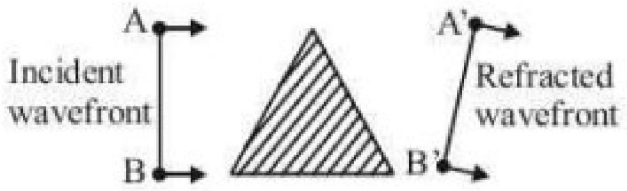
A) A is correct but R is correct

B) A is not correct but R is correct

C) Both A and R are correct and R is the correct explanation of A

D) Both A and R are correct and R is not the correct explanation of A

7. Figure shows behavior of a wavefront when it passes through a prism.



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Which of the following statement(s) is/are correct.

(I) Lower portion of wavefront (B') is delayed resulting in a tilt.

(II) Time taken by light to reach A to A' is equal to that of from B to B'.

(III) Speed of wavefront is same ever where.

(IV) Points on wavefront A'B' is in phase with points on wavefront AB.

A) I and II only B) II and III only C) III and IV only D) I and III only

8. Given below are two statements:

Statement I: When a capillary tube is dipped into a liquid, the liquid neither rises nor falls in the capillary. The contact angle may be 0^0.

Statement II: The contact angle between a solid and a liquid is a property of the material of the solid and liquid as well.

In the light of above statement, choose the correct answer from the options given below

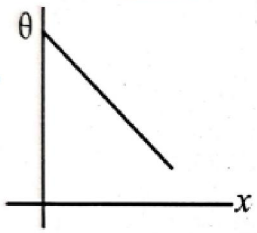
A) Statement I is false, Statement II is true

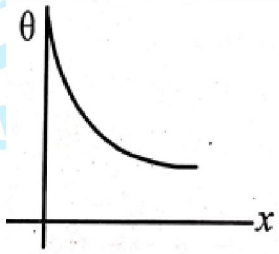
B) Both Statement I and statement II are true

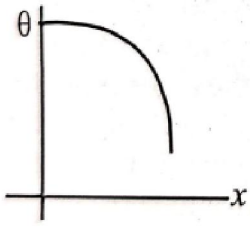
C) Both Statement I and statement II are false

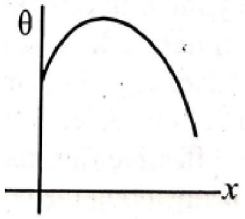
D) Statement I is true, Statement II is false

9. A long metallic bar of uniform cross section is carrying heat from one of its ends to the other end under steady-state. The variation of temperature θ along the length x of the bar from its hot end is best described by which of the following figures.

A) 

B) 

C) 

D) 

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10. If mass-energy equivalence is taken into account, when water is cooled to form ice, the mass of water should

A) increase B) remain unchanged

C) decrease D) first increase then decrease

11. A paramagnetic material is placed in a magnetic field. Consider the following statements:

(A) If the magnetic field is increases, the magnetization is increased

(B) If the temperature is increased, the magnetization is increased.

A) Both A and B are true B) A is true but B is false

C) B is true but A is false D) Both A and B are false

12. Electromagnets are made of soft iron because soft iron has

A) high retentivity and high coercive force

B) high retentivity and low coercive force

C) low retentivity and high coercive force

D) low retentivity and low coercive force

13. Two bodies at different temperatures are mixed in a calorimeter. Which of the following quantities remains conserved (Water equivalent of calorimeter is zero)

A) Sum of the temperatures of the two bodies

B) Total heat of the two bodies

C) Total internal energy of the two bodies

D) Internal energy of each body

14. A closed cubical box is completely filled with water and is accelerated horizontally towards right with an acceleration a. The resultant normal force by the water on the top of the box.

A) passes through the centre of the top

B) passes through a point to the right of the centre

C) passes through a point to the left of the centre

D) becomes zero

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15. Suppose a container is evacuated to leave just one molecule of a gas in it. Let υ_a *and* υ_{rms} represent the average speed and the rms speed of the gas.

A) $\upsilon_a > \upsilon_{rms}$ B) $\upsilon_a = \upsilon_{rms}$ C) $\upsilon_a < \upsilon_{rms}$ D) υ_{rms} is undefined

16. If the source of light used in Young's double slit experiment is changed from red to violet,

A) the fringes will become brighter

B) consecutive fringes will come closer

C) the intensity of minima will increase

D) the central bright fringe will become a dark fringe

17. A hole diffuses from the p-side and n-side in a p-n junction. This mean that

A) a bond is broken on the n-side and the electron freed from the bond jumps to the conduction band

B) a bond is broken on the n-side and the electron freed from the bond jumps to a broken bond on the p-side to complete it

C) a conduction electron on the p-side jumps to a broken bond to complete it

D) a bond is broken on the p-side and the electron freed from the bond jumps to a broken bond on the n-side to complete it.

18. The drift current in a $p-n$ junction is

A) from the n-side to the p-side

B) from the p-side to the n-side

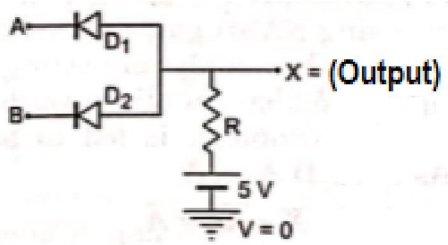
C) from the n-side to the p-side if the junction is forward-biased and in the opposite direction if it is reversed biased

D) from the p-side to the $n-$side if the junction is forward-biased and in the opposite direction if it is reversed-biased.

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19. The circuit given below represents which of the logic operations (A and B are considered ON if $V_A = V_B = 5V$)



A) AND B) NOT C) OR D) NOR

20. In case of linearly polarized light, the magnitude of the electric field vector

A) does not change with time

B) varies periodically with time

C) increases and decreases linearly with time

D) is parallel to the direction of propagation

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

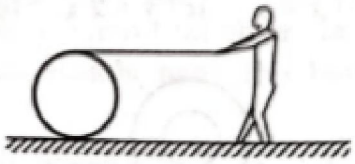
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
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21. A body of mass 'm' is projected with a speed 'u' making an angle of 45^0 with the ground. The angular momentum of the body about the point of projection, at the highest point is expressed as $\frac{\sqrt{2}mu^3}{Xg}$. The value of 'X' is _____

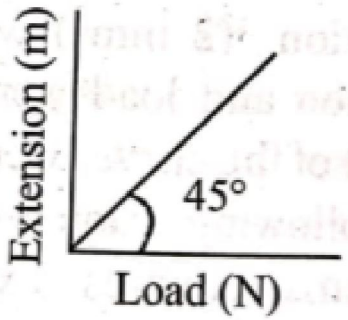
22. A string of negligible thickness is wrapped several times around a cylinder kept on a rough horizontal surface. A man standing at a distance l from the cylinder hold one end of the string and pulls the cylinder towards him. There is no slipping anywhere. The length of the string passed through the hand of the man while the cylinder reaches his hands is is x l . Then x =

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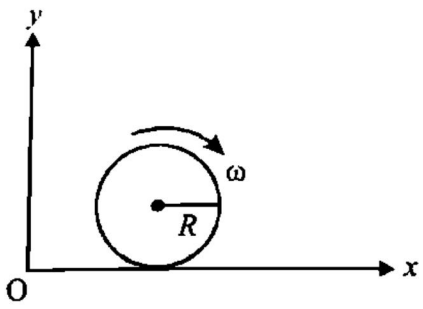


23. As shown in the figure, in an experiment to determine Young's modulus of a wire, the extension-load curve is plotted. The curve is a straight line passing through the origin and makes an angle of 45^0 with the load axis. The length of wire is 62.8 cm and its diameter is 4mm. The Young's modulus is found to be $x \times 10^4\ Nm^{-2}$. The value of x is



24. A cylindrical tube, open at both ends, has a fundamental frequency v. The tube is dipped vertically in water so that half of its length is inside the water. The new fundamental frequency is $\frac{2v}{x}$, then $x =$____

25. A spherical shell of 1 kg mass and radius R is rolling with angular speed ω on horizontal plane (as shown in figure). The magnitude of angular momentum of the shell about the origin O is $\frac{a}{3}R^2\omega$. The value of 'a' will be:



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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Choose the **incorrect** statement regarding brown ring test:

A) Freshly prepared ferrous sulphate is used to avoid aerial oxidation of Fe^{2+} into Fe^{3+}

B) Ring is formed due to nitroso ferrous sulphate complex

C) Spin only magnetic moment of the brown ring complex is $\sqrt{15}$ BM.

D) Concentrated nitric acid is used in the test.

27. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion(A): Benzene upon reaction with chlorine in presence of UV light at 500 K forms

Benzene hexachloride.

Reason(R): The reaction follows free radical addition mechanism.

In the light of the above statements, choose the correct answer from the options given below.

A) (A) is true but (R) is false

B) Both (A) and (R) are true and (R) is the correct explanation of (A)

C) (A) is false but (R) is true

D) Both (A) and (R) are true but (R) is not the correct explanation of (A)

28. A chloride salt of transition metal does not give any precipitate with acidified H_2S but it gives a black precipitate with ammonical H_2S. The aqueous solution of the salt gives brilliant red precipitate with ammonical dimethyl glyoxime. The colour produced by the borax bead of the salt in hot non-luminous flame is:

A) Grey B) Violet C) Reddish brown D) Blue

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29. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Propene on reaction with HBr in presence of peroxide gives Anti Markonikov's product while with HCl under similar conditions gives Markonikov's product

Reason(R): HCl is more polar than HBr

A) (A) is true but (R) is false

B) Both (A) and (R) are true and (R) is the correct explanation of (A)

C) (A) is false but (R) is true

D) Both (A) and (R) are true but (R) is not the correct explanation of (A)

30. Choose the correct statement(s):

i) BF_3 and $AlCl_3$ are electron deficient compounds.

ii) HCl and NH_3 are octet molecules.

iii) SF_4 has folded square shape.

iv) Bond angle in $SnCl_2$ is more than that of SO_2

A) i and ii only B) i, ii and iii only

C) ii and iv only D) All

31. If s, p_x and p_y orbitals are involved in hybridization of sulphur in SO_2. Then its $d_\pi - p_\pi$ bond is formed by:

A) d_{z^2} and p_z B) d_{xy} and p_x C) d_{yz} and p_x D) d_{yz} and p_z

32. Identify the incorrect statement:

A) Net work done in reversible isothermal cyclic process is zero

B) Net work done in irreversible isothermal cyclic process is non-zero

C) Magnitude of work done is more in case of adiabatic reversible compression than isothermal reversible compression for the same volume change.

D) Magnitude of work done in reversible isobaric expansion is greater than irreversible isobaric expansion.

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33. Identify the correct statement/s:

1) Work and heat are the only forms of energy that are exchanged between system and surroundings during a process.

2) Entropy change of surroundings takes place only due to heat exchange between system and surroundings.

3) Increase of entropy of system associated with decrease of enthalpy of system always the process is spontaneous.

4) In isothermal reversible process, the entropy change of system is directly proportional to heat exchanged between system and surroundings.

A) 1,2,3 B) 1,2,4 C) 1,3,4 D) 1,2,3,4

34. Identify the incorrect statement related to quantum mechanical model of atom?

A) The solution gives the possible energy levels the electron can occupy and the corresponding wave function(s) (ψ) of the electron associated with each energy level.

B) When quantum mechanics is applied to macroscopic objects (for which wave like properties are insignificant) the results are the same as those from the classical mechanics.

C) The wave function is a mathematical function whose value depends upon the coordinates of the electron in the atom and does not carry any physical meaning.

D) The probability of finding an electron at a point within an atom is proportional to the $|\psi|^2$ in that region.

35. Which of the following statement is not true according to particle nature of radiation?

A) The absorption and emission of radiation arises from oscillator i.e., atoms in the wall of black body.

B) The number of electrons ejected is proportional to the intensity or brightness of light in photo electric effect.

C) Intensities of radiations of different wavelengths emitted by hot body depend upon its temperature.

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D) The kinetic energies of photo electrons increase with the increase of frequency of the light used.

36. Find out the incorrect statements according to collision theory:

A) The reactant molecules are assumed to be hard spheres and reaction is postulated to occur when molecules collide with each other.

B) The number of collisions per second per unit volume of the reaction mixture is known as collision frequency.

C) Arrhenius constant is related to collision frequency.

D) $Rate = Z_{AB}\, e^{-E_a/RT}$ is the equation that predicts the rate constant fairly for all types of reactions.

37. Correct statement among the following is:

A) With increase of temperature, rate of exothermic reaction decreases while that of endothermic reaction increases.

B) Zero order reaction never goes to completion practically.

C) Arrhenius theory is strictly applicable to first order reactions only.

D) A catalyst can initiate a non-spontaneous reaction also.

38. IE_2 for an element is invariable higher than IE_1 because:

A) It is difficult to remove electron from cation

B) The size of the cation is smaller than its atom

C) Z_{eff} is more for cation

D) All the above

39. Which of the following is used as a spraying reagent in paper chromatography?

A) Concentrated hydrochloric acid solution

B) Sodium chloride solution

C) Ninhydrin solution

D) Copper sulphate solution

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40. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Dipole moment of NF_3 is less than NH_3

Reason(R): In NH_3, the orbital dipole due to lone pair and the dipole moment of N-H bonds are in same direction, but in NF_3 the orbital dipole due to lone pair and dipole moments of N-F bonds are in opposite direction.

In the light of the above statements, choose the correct answer from the options given below.

A) (A) is true but (R) is false

B) Both (A) and (R) are true and (R) is the correct explanation of (A)

C) (A) is false but (R) is true

D) Both (A) and (R) are true but (R) is not the correct explanation of (A).

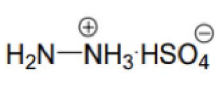
41. The correct match between item I and II is

Item-I(Mixture)	Item-II(Separation method)
(A) H_2O : Sugar	(P) Sublimation
(B) H_2O : Aniline	(Q) Recrystallization
(C) H_2O : Toulene	(R) Steam distillation
	(S) Differential extraction

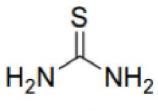
A) A-Q; B-R; C-S B) A-Q; B-R; C-P

C) A-S; B-R; C-P D) A-R; B-P; C-S

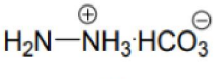
42. The compounds those would not respond to tests of both nitrogen and sulfur with sodium fusion extracts are:



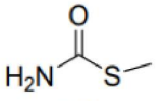
I



II



III



IV

A) I and III B) III and IV C) I and IV D) II and IV

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43. Find the incorrect order of acidic strength -

A) $CCl_3COOH > CHCl_2COOH > NO_2CH_2COOH > NC-CH_2COOH$

B) $NO_2CH_2COOH > F.CH_2COOH > Cl\cdot CH_2COOH > HCOOH$

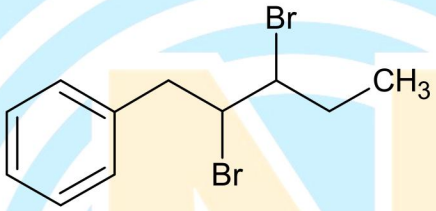
C) $HCOOH > C_6H_5COOH > C_6H_5CH_2COOH > CH_3COOH$

D) $CF_3COOH > NO_2\cdot CH_2COOH > NC\cdot CH_2COOH > CHCl_2COOH$

44. Alkanes show free radical substitution reaction except with:

A) H_2SO_4 B) HNO_3 C) Cl_2 D) V_2O_5

45. The correct IUPAC name of the following compound is:



A) 2,3-dibromopentylbenzene B) 2,3-dibromo-1-phenyl pentane

C) 1-phenyl-2,3-dibromopentane D) 3,4-dibromopentylbenzene

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
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46. Among the following, the number of correct statements is_____.

i) Alkenes are easily attacked by electrophilic reagents due to presence of loosely held π- electrons.

ii) Alkenes are unstable molecules in comparison to alkanes due to presence of weaker π-bond

iii) Ethanol can form any of the ethyl hydrogen sulphate, ethene or diethyl ether on reaction with conc. Sulphuricacid under appropriate conditions.

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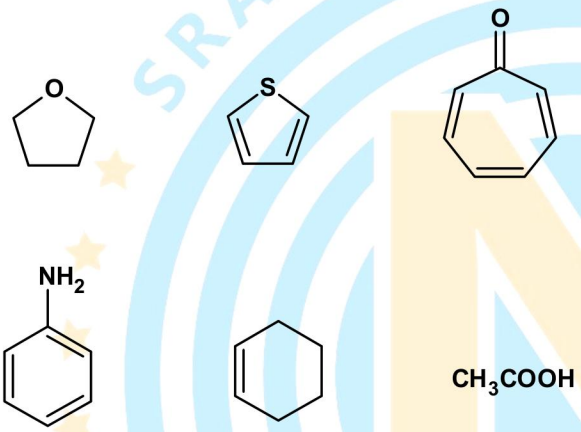
iv) Ozonolysis reaction is highly useful in detecting the position of the double bond in alkenes or other unsaturated compounds.

v) Thedelocalized π electron cloud of benzene is attracted more strongly by the nuclei of the carbon atoms than the electron cloud localized between two carbon atoms.

vi) Benzene and polynuclear hydrocarbons containing more than two benzene rings fused together are toxic and carcinogenic.

47. If number of elements in 5th period of periodic table is equal to x and number of elements in the 6th period of periodic table is equal to y then find the value of (x + y) ?

48. The total number of alicyclic compounds among the following is_____



49. 2.2 g of nitrous oxide (N_2O) gas is cooled at a constant pressure of 1 atm from 310 K to 270 K causing the compression of the gas from 217.1 mL to 167.75 mL. The change in internal energy of the process, ΔU is '-x' J. The value of 'x' is [nearest integer]
(Given: atomic mass of N = 14 g mol^{-1} and of O = 16 g mol^{-1}. Molar heat capacity of N_2O is 100 JK^{-1} mol^{-1}, 1 L-atm=101.3J)

50. For a pseudo first order reaction. Rate constant is represented by

$\log k\left(s^{-1}\right)=15-\frac{9.19\times10^{3}(K)}{T}$, then the temperature (in kelvin) at which its half life is 28min is_____ $\left(ln2=0.7, log3=0.48, log2=0.30\right)$

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If a is the AM of b and c and two GMS G_1 and G_2 are inserted between b and c, then $G_1^3 + G_2^3 =$

(A) abc (B) 2abc (C) 3abc (D) 4abc

52. The minimum value of $a\sec^2 x + b\,\mathrm{cosec}^2 x$ is (where $a, b > 0$)

(A) $(\sqrt{a} - \sqrt{b})^2$ (B) $|a - b|$ (C) $(\sqrt{a} + \sqrt{b})^2$ (D) $a + b$

53. $|z_1 + z_2|^2 + |z_1 - z_2|^2 =$

(A) $|z_1|^2 + |z_2|^2$ (B) $|z_1|^2 - |z_2|^2$ (C) $2(|z_1|^2 + |z_2|^2)$ (D) $2(|z_1|^2 - |z_2|^2)$

54. Let z_1, z_2 be two fixed points in the Argand plane and z be a variable complex number satisfying $\left|\frac{z - z_1}{z - z_2}\right| = 3$, then the locus of z is a circle with

(A) z_1 as an interior point (B) z_2 as an interior point

(C) z_1 and z_2 as interior points (D) None of z_1, z_2 be in the interior

55. If $x_r = \cos\left(\frac{\pi}{2^r}\right) + i\sin\left(\frac{\pi}{2^r}\right)$ $r = 1, 2, 3, \ldots$, then $\prod_{r=1}^{\infty} x_r = ?$

(A) i (B) – i (C) 1 (D) –1

56. The value of ${}^{12}C_0 - {}^{12}C_2 + {}^{12}C_4 \ldots + {}^{12}C_{12} = ?$

(A) 32 (B) –32 (C) 64 (D) –64

57. The value of $\binom{30}{0}\binom{30}{20} - \binom{30}{1}\binom{30}{19} + \binom{30}{2}\binom{30}{18} - \ldots + \binom{30}{20}\binom{30}{0} = ?$

(A) $\binom{30}{20}$ (B) $-\binom{30}{10}$ (C) 0 (D) $\binom{30}{30}$

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58. The number of ways of distributing 10 identical fruits amongst 4 boys such that any boy may secure any number of fruits and not all fruits need to be distributed is

(A) ${}^{13}C_3$ (B) ${}^{13}C_4$ (C) ${}^{14}C_4$ (D) ${}^{14}C_5$

59. Let D_n denote the number of ways of derangement of n symbols, then

${}^7C_0.D_7 + {}^7C_1.D_6 + ... + {}^7C_7.D_0 = ?$ (assume $D_0 = 1$)

(A) 120 (B) 720 (C) 5040 (D) 40320

60. If the equation $ax^2 + bx + c = 0$ has imaginary roots and $a - b + c < 0$ then which of the following is true?

(A) $a + b + c > 0$ (B) $4a + 2b + c > 0$ (C) $a + b + c < 0$ (D) $a + 2b < 0$

61. If $y = mx + c$ is a tangent to $y^2 = -8ax$, then

(A) $c = \frac{a}{m}$ (B) $c = \frac{2a}{m}$ (C) $c = -\frac{2a}{m}$ (D) $c = -\frac{a}{m}$

62. If the circles $x^2 + y^2 + g_1x + f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ intersect orthogonally, then

(A) $2(g_1g_2 + f_1f_2) = c_1 + c_2$ (B) $g_1g_2 + f_1f_2 = c_1 + c_2$

(C) $g_1g_2 + f_1f_2 = 2(c_1 + c_2)$ (D) $g_1g_2 + 2f_1f_2 = c_1 + c_2$

63. Let P be a variable point on the ellipse $ax^2 + by^2 = 1$ (where $0 < a < b$). If S_1 and S_2 are the foci, then $PS_1 + PS_2 =$

(A) $\frac{2}{a}$ (B) $\frac{2}{b}$ (C) $\frac{2}{\sqrt{a}}$ (D) $\frac{2}{\sqrt{b}}$

64. From the foci of the hyperbola $4x^2 - 9y^2 = 1$, perpendicular are dropped onto the tangent at the point $\left(1, \frac{1}{\sqrt{3}}\right)$. The product of the perpendicular is equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{9}$

65. A variable line L is such that the algebraic sum of its distances from the points (1, 2), (3, –5) and (8, –9) is zero. Then, the line L must pass through the point

(A) (0, 0) (B) (4, –4) (C) (6, –2) (D) (–3, 4)

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66. Equation of the radical axis of the circles $x^2+y^2+2x+y-3=0$ and $2x^2+2y^2+x-4y+6=0$ is

(A) $x^2+y^2-x-5y+9=0$ (B) $x+2y-4=0$

(C) $2x+y+3=0$ (D) $x-y-3=0$

67. Equation of the chord joining the points $P\left(\frac{at_1}{2},\frac{at_1^2}{4}\right)$ and $Q\left(\frac{at_2}{2},\frac{at_2^2}{4}\right)$ on the parabola $x^2=ay$ is

(A) $2x(t_1+t_2)=4y+at_1t_2$ (B) $y(t_1+t_2)=2x+2at_1t_2$

(C) $x(t_1+t_2)=2y+at_1t_2$ (D) $y(t_1+t_2)=\frac{x}{2}+at_1t_2$

68. The value of $\left({}^{30}C_0\right)^2-\left({}^{30}C_1\right)^2+\left({}^{30}C_2\right)^2-....+\left({}^{30}C_{30}\right)^2=$

(A) ${}^{60}C_{30}$ (B) ${}^{30}C_{15}$ (C) $-{}^{30}C_{15}$ (D) 0

69. The no. of ways to divide 20 distinct objects into 5 unordered groups, each containing 4 objects is =

(A) $\frac{(20)!}{(120)^4}$ (B) $\frac{(20)!}{(24)^5}$ (C) $\frac{(20)!}{(120)^4(24)}$ (D) $\frac{(19)!}{6(24)^5}$

70. A variable tangent to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ meets the coordinate axes at points A and B. The minimum possible area of ΔAOB is equal to

(A) 2 (B) 4 (C) 6 (D) 8

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:

Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Maximum possible distance of the point (1,3) from a variable point on the circle $x^2+y^2-2x+8y+1=0$ is equal to

72. If perpendicular tangents can be drawn from the point $P(2,k)$ to the hyperbola $\frac{y^2}{16}-\frac{x^2}{4}=2$, then $k^2=$

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73. The angle between the asymptotes of the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ containing focus is equal to $\tan^{-1}\left(\frac{a}{b}\right)$, where a, b are coprime positive integers. The value of $a+b=$

74. Let $a_k = e^{i\frac{k\pi}{20}}$, $k \in N$. The value of $\sum_{k=1}^{40}(a_k^{80})+\left(\sum_{k=1}^{39}a_k\right)^{79} =$

75. If $y=2x-\lambda$ is a normal to the parabola $(y-2)^2=8(x-3)$, then $\lambda =$

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